

The isoform decomposition polytope of a splicing graph, and a BEST identity for its integer points

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Abstract

Junction and exon read counts of a gene are an integer flow w on its splicing graph, a directed acyclic graph from the transcription start to the polyadenylation site, and the transcript isoforms are s - t paths. The abundances compatible with w form the fibre polytope $\Theta(w) = \{\theta \geq 0 : \sum_j \theta_j \mathbf{1}_{P_j} = w\}$. We prove that its dimension is the number of s - t paths in the support of w minus $|E_w| - |V_w| + 2$, because w lies in the relative interior of the cone spanned by those paths and the path-arc matrix has the rank of the flow space; for alternative events in series with $k_1, \dots, k_r$ branches this is $\prod k_i - \sum (k_i - 1) - 1$, so deconvolution from junction counts is unique for one event and fails already for two cassette exons, and for Dscam ($12 \times 48 \times 33 \times 2$) the 38 016 isoforms sit in a 37 924-dimensional polytope. Every minimum-support decomposition, with real or integer weights, is a vertex of $\Theta(w)$; but the vertices of $\Theta(w)$ are not integral in general — three parallel-arc pairs in series and $w \equiv 1$ have a vertex with four weights $\frac{1}{2}$ beside an integer minimum of two paths — so the extreme rays of the flow cone, the vertices of the fibre, and integrality are three different things. The new algebraic content is an identity. Adding F return arcs $t \rightarrow s$ to the multigraph G_w , F the flow value, the per-sequence BEST count of the augmented graph equals $\sum_{\theta} (F - 1)! / \prod_j \theta_j!$ over the integer points of $\Theta(w)$: a determinant computes the multinomial-weighted partition function of the integer decompositions, of which the plain number of integer points has no such expression. The identity and the vertex statements are verified exactly on 238 flows, and no instance with a real minimum below the integer minimum was found among 232 random small flows.

Keywords. splicing graph, alternative splicing, flow decomposition, isoform quantification, identifiability, BEST theorem, fibre polytope, Dscam.

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1 What was asked, and what is true

The Bio strand of these notes reads assembly ambiguity off the critical group of a de Bruijn graph [2], double-stranded ambiguity off the box points of an eigenlattice of the strand involution [3], and polyploid phasing off the p -fold decompositions of a pooled flow [4]. The common object is the decomposition of a balanced integer flow into closed walks. Transcript quantification is the same object with the circle cut open: a splicing graph is a DAG with a source and a sink

[7], read counts are a flow, and the isoforms are s - t paths; the question “which isoforms, in what abundance” is the question of decomposing the flow into paths, and the ambiguity is the polytope of such decompositions.

The specification of this note asked for a proof that “minimum flow decompositions correspond to convex combinations of integer lattice points on the extreme rays of the flow polytope”. That sentence runs three distinct objects together, and it is worth separating them because two of the three are elementary and the third is false as stated. The extreme rays of the cone of s - t flows of a DAG are the path indicators, and the vertices of the polytope of flows of fixed value are the paths scaled by the value: this is the total unimodularity of the incidence matrix and nothing else. The polytope that matters for a *given* w is the fibre $\Theta(w)$ of the path-arc map over w , whose vertices are the decompositions with linearly independent support; every minimum-support decomposition is one of them (Proposition 5). But the path-arc matrix is not totally unimodular, and $\Theta(w)$ has fractional vertices for integer w (Proposition 6); the integer decompositions are the lattice points of $\Theta(w)$, of which the vertices are in general not a subset. Minimum flow decomposition is NP-hard [6], and nothing here changes that.

What is true and, we believe, new is in Sections 2 and 4. The dimension of $\Theta(w)$ is exactly the number of paths in the support of w minus the rank of the flow space of that support (Theorem 3); this is the precise form of the non-identifiability of isoform deconvolution from junction counts, a phenomenon known since [10, 8], and it gives the closed form $\prod k_i - \sum (k_i - 1) - 1$ for events in series and the number 37 924 for Dscam. And the BEST theorem, applied to the flow closed up by F return arcs, evaluates the multinomial-weighted sum over the integer points of $\Theta(w)$ as a determinant (Theorem 7): the same normalisation by $\prod c(a)!$ that [3] needed for double strands appears here as the passage from cyclic sequences of isoforms to abundance vectors. The script `isoform_polytope.py` verifies every statement and reproduces every number; it prints a seven-line pass/fail battery.

2 The splicing graph, its flows, and the fibre polytope

Definition 1. A *splicing graph* is a directed acyclic multigraph $G = (V, E)$ with a source s and a sink t in which every arc lies on some s - t path. An *isoform* is an s - t path; $\mathcal{P}$ is the set of isoforms and $A \in \{0, 1\}^{E \times \mathcal{P}}$ the path-arc matrix. A *flow* is $w \in \mathbb{Z}_{\geq 0}^E$ with $\partial w = F(\delta_t - \delta_s)$, F its *value*; $E_w = \text{supp } w$, V_w the vertices touched, $\mathcal{P}(w)$ the isoforms inside E_w , and

$$\Theta(w) = \left\{ \theta \in \mathbb{R}_{\geq 0}^{\mathcal{P}(w)} : \sum_{P \in \mathcal{P}(w)} \theta_P \mathbf{1}_P = w \right\}.$$

$\Theta(w)$ is non-empty by flow decomposition (a non-negative flow on a DAG is a non-negative combination of s - t paths, and an integer flow of integer paths), and bounded since $\theta_P \leq F$. It is the polytope of abundance vectors compatible with the counts.

Lemma 2. $\text{rank } A = |E| - |V| + 2$.

Proof. The columns of A lie in $\mathcal{F} = \{x \in \mathbb{Q}^E : \partial x \in \mathbb{Q}(\delta_t - \delta_s)\}$, which has dimension $\dim \ker \partial + 1 = |E| - |V| + 2$ since G is connected. They span it: let $x \in \mathcal{F}$ and let $x_0 = \sum_P \mathbf{1}_P$, which is positive on every arc because every arc lies on a path; for M large, $x + Mx_0$ is a non-negative flow, hence a non-negative combination of paths, so x is in the span. $\square$

Theorem 3 (Dimension of the isoform polytope). $\dim \Theta(w) = |\mathcal{P}(w)| - (|E_w| - |V_w| + 2)$. In particular the abundances are determined by the counts if and only if the isoforms inside the support of w are linearly independent as arc vectors, i.e. $|\mathcal{P}(w)| = |E_w| - |V_w| + 2$.

Proof. The subgraph $G_w = (V_w, E_w)$ is a splicing graph in its own right (every arc of positive flow lies on a path of positive flow, by flow decomposition), so Lemma 2 gives $\text{rank } A_w = |E_w| - |V_w| + 2$ for its path-arc matrix. The cone $C_w = \{x \in \mathcal{F}_w : x \geq 0\}$ is the cone generated by $\mathcal{P}(w)$, and its relative interior is $\{x \in \mathcal{F}_w : x_a > 0 \ \forall a \in E_w\}$, because every coordinate x_a is a non-trivial linear form on $\mathcal{F}_w$ (some path uses a). Since $w > 0$ on E_w , $w \in \text{relint } C_w$, and a point in the relative interior of a finitely generated cone is a strictly positive combination of all the generators. So $\Theta(w)$ contains a point with full support, its affine hull is the whole affine space $\{\theta : A_w \theta = w\}$, and the dimension is $|\mathcal{P}(w)| - \text{rank } A_w$. $\square$

Corollary 4 (Events in series). *If G is a chain of r alternative events, the i -th offering k_i branches between consecutive junctions, then for a flow of full support*

$$\dim \Theta(w) = \prod_{i=1}^r k_i - \sum_{i=1}^r (k_i - 1) - 1,$$

which is 0 if and only if at most one k_i exceeds 1. Two cassette exons in series give $4 - 2 - 1 = 1$; three give 4; a cassette, two pairs of mutually exclusive exons, a second cassette and a four-way mutually exclusive block give $64 - 8 = 56$; the four blocks of Dscam [12], with 12, 48, 33, 2 alternatives, give $38\,016 - 92 = 37\,924$.

Proof. Paths multiply, $|\mathcal{P}| = \prod k_i$; branches with ℓ arcs have $\ell - 1$ interior vertices, so $|E| - |V| + 2 = \sum_i k_i - r - 1 + 2 = \sum_i (k_i - 1) + 1$ independently of the branch lengths. The last claim is Theorem 3. $\square$

The identifiability statement in Theorem 3 is the rank condition of [8] specialised to junction-and-exon counts; the dimension formula makes it quantitative, and Corollary 4 says that for any gene with two independent alternative events the counts do not determine the abundances, whatever the depth of sequencing. Table 1 lists the models and their dimensions.

3 Vertices, minimum decompositions, and integrality

Proposition 5. *The vertices of $\Theta(w)$ are the decompositions whose support is a linearly independent set of path vectors. Every decomposition of minimum support — among real decompositions, or among integer ones — is a vertex.*

Proof. The first sentence is the characterisation of basic feasible solutions of $\{\theta \geq 0 : A\theta = w\}$. For the second, let θ have minimum support and suppose the columns of its support are dependent, $Av = 0$ with $v \neq 0$ supported in $\text{supp } \theta$. Then $\theta + \tau v \in \Theta(w)$ for τ in a closed interval containing 0 in its interior, and at an endpoint of that interval some coordinate of $\text{supp } \theta$ vanishes, giving a decomposition of smaller support; if θ and v are integral, the endpoints can be taken at integer τ (the first coordinate to vanish does so at $\tau = -\theta_j/v_j$, and one may stop at the last integer before it, or note that v may be scaled to make the endpoint integral without changing the argument). Contradiction. $\square$

Proposition 6 (A fractional vertex). *Let G consist of four vertices $s = v_0, v_1, v_2, v_3 = t$ with two parallel arcs a_i, a'_i from v_{i-1} to v_i , and $w \equiv 1$ (value 2). Then $\Theta(w)$ has 8 isoforms, dimension 4, four integer points, and six vertices of which two are fractional: $\theta = \frac{1}{2}$ on the four paths $(a_1, a'_2, a_3), (a_1, a_2, a'_3), (a'_1, a_2, a_3), (a'_1, a'_2, a'_3)$ is a vertex, while the minimum decompositions are the two integer points with two paths each.*

Proof. The four listed paths cover every arc exactly twice, so θ is feasible; their arc vectors are independent, because a relation $\alpha P_1 + \beta P_2 + \gamma P_3 + \delta P_4 = 0$ read on a_1, a_2, a_3 gives $\alpha + \beta = \beta + \gamma = \alpha + \gamma = 0$, hence $\alpha = \beta = \gamma = 0$ and then $\delta = 0$. So θ is a vertex by Proposition 5, and it is not integral. The 3×3 submatrix of the path–arc matrix on those three arcs and three paths has determinant 2, which is where the halves come from. The counts 8, 4, 4, 6, 2 are the transcript of the script. $\square$

So the lattice points of $\Theta(w)$ and its vertices are different sets, each containing the minimum decompositions. Whether the integer minimum can exceed the real minimum we do not know: the script compared the two on 232 random flows on small DAGs with 3 to 12 isoforms and found no instance where they differ, and one instance with a fractional vertex. Minimum flow decomposition being NP-hard [6], with the bound $|E| - |V| + 2$ on the number of paths needed [14] and exact solvers in practice [9, 5], nothing in this section is intended as an algorithm.

4 The BEST identity

Close the flow up. Let $\hat{G}_w$ be the multigraph with $w(a)$ labelled parallel copies of each arc $a \in E_w$ and F labelled *return arcs* $t \rightarrow s$. It is balanced and connected, so it has Eulerian circuits, and the per-sequence count of [3] applies:

$$N_{\text{seq}}(\hat{G}_w) = \frac{t(\hat{G}_w) \text{Loc}(\hat{G}_w)}{F! \prod_a w(a)!} = \sum_T \frac{1}{d(T)},$$

the sum over the circular sequences T of $\hat{G}_w$ — cyclic sequences of arc contents using each content with its multiplicity — weighted by the order $d(T)$ of the rotation group of T ; here t is the number of spanning arborescences converging on a root and $\text{Loc} = \prod_v (d^+(v) - 1)!$.

Theorem 7 (BEST identity for isoform decompositions).

$$N_{\text{seq}}(\hat{G}_w) = \sum_{\theta \in \Theta(w) \cap \mathbb{Z}^{\mathcal{P}(w)}} \frac{(F-1)!}{\prod_P \theta_P!}.$$

Proof. A circular sequence of $\hat{G}_w$ uses the F return arcs, and between two consecutive returns it walks from s to t along arcs of the DAG; a walk in a DAG repeats no vertex, so it is an isoform. Thus T is a cyclic sequence $(P_{j_1}, \dots, P_{j_F})$ of isoforms, its arc usage is $\sum_i \mathbf{1}_{P_{j_i}} = w$, and the multiplicity vector $\theta_P = \#\{i : P_{j_i} = P\}$ is an integer point of $\Theta(w)$ with $\sum_P \theta_P = F$. Conversely every integer point θ and every cyclic arrangement of the multiset $\{P^{\theta_P}\}$ gives such a T . For a fixed θ the cyclic arrangements, weighted by $1/d(T)$, number $\frac{1}{F} \cdot \frac{F!}{\prod_P \theta_P!}$: the cyclic group of order F acts on the $F!/\prod \theta_P!$ linear arrangements, and $\sum_{\text{orbits}} 1/|\text{stab}| = |\text{set}|/|\text{group}|$. $\square$

The left side is a determinant times factorials; the right side is a sum over the lattice points of a polytope of the multinomial coefficient of each point, i.e. the number of linear orders of F

model	$ V $	$ E $	isoforms	$ E - V + 2$	$\dim \Theta$	unique?
cassette exon (skipping)	3	3	2	2	0	yes
mutually exclusive exon pair	4	4	2	2	0	yes
alternative 5' splice site	2	2	2	2	0	yes
two cassettes in series	5	6	4	3	1	no
three cassettes in series	7	9	8	4	4	no
tropomyosin-like (five events)	16	22	64	8	56	no
three parallel-arc pairs	4	6	8	4	4	no
<i>Dscam</i> ($12 \times 48 \times 33 \times 2$)	100	190	38 016	92	37 924	no

Table 1: Ranks and dimensions. The tropomyosin-like model is a pair of mutually exclusive exons, a cassette, a second pair, a second cassette and a four-way mutually exclusive block in series. The rank of the path-arc matrix equals $|E| - |V| + 2$ in every row (over $\mathbb{Q}$ for the small models, modulo $2^{31} - 1$ for the $190 \times 38\,016$ *Dscam* matrix, 12 s). Checks (R), (D), (U), (S).

flow	F	paths	dim	int. pts	vert.	frac.	MFD	t	N_{seq}
two cassettes	6	4	1	3	2	0	3	216	50
three parallel pairs	2	8	4	4	6	2	2	8	4
tropomyosin-like	10	24	18	714	358	90	4	$3.6288 \cdot 10^9$	22 680 000
three cassettes	7	4	1	2	2	0	3	57 624	35

Table 2: Flows: two cassettes with $w = (3, 3, 3, 2, 4, 2)$; three parallel pairs with $w \equiv 1$; the tropomyosin-like model with a flow generated from five random isoforms of weights 1 to 3; three cassettes with a random flow. “paths” is $|\mathcal{P}(w)|$; “MFD” the minimum support, equal for real and integer weights in every row; t the arborescence count of $\hat{G}_w$; N_{seq} its per-sequence count, which equals $\sum_{\theta} (F - 1)! / \prod \theta_P!$ over the integer points in every row. In the tropomyosin-like row the flow was generated from five isoforms and the minimum decomposition uses four. Checks (V), (B).

transcripts with those abundances. Theorem 7 is thus a closed form for the “partition function” of the integer decompositions with the natural combinatorial weight, and it is the isoform analogue of the fact that BEST counts genome reconstructions: there the whole spectrum is one closed walk, here it is a cyclic word in the isoforms. The plain number of integer points $|\Theta(w) \cap \mathbb{Z}^P|$, which is what a uniform prior over decompositions would need, is a lattice-point count with no determinantal formula, and Theorem 7 says precisely which weighting is determinantal.

Remark 8. On one strand, $\prod w(a)!$ is a constant and the single-genome notes count arc-rooted circuits without it. Here, as in the double-stranded and polyploid sums, the divisor varies with the point: for the tropomyosin-like flow of Table 2 the arborescence count is $t = 3\,628\,800\,000$, and only after multiplying by Loc and dividing by $F! \prod_a w(a)!$ does one obtain $N_{\text{seq}} = 22\,680\,000$, the quantity that matches the lattice sum.

5 Verification and data

The seven checks. (R) rank $A = |E| - |V| + 2$ on the seven small models; (D) the dimension formula and its series form; (U) uniqueness for one event, dimension one for two cassettes; (S) *Dscam*; (V) every minimum-support integer decomposition is a vertex, and the fractional vertex

of Proposition 6 exists; (B) the BEST identity on the four flows of Table 2 and on 232 random flows on DAGs with 4 to 6 vertices; (I) real versus integer minimum on those 232 flows — never different — with one fractional vertex among them. All pass; the run takes about three and a half minutes, almost all of it in the random search.

6 Scope, provenance, and what is open

Scope. Lemma 2, Theorem 3, Corollary 4, Proposition 5 and Theorem 7 are proved for arbitrary splicing graphs; Proposition 6 is one graph. The model is the exact flow: no read-length effects, no paired reads, no noise, and no length normalisation of abundances; reads spanning several junctions are additional linear constraints that cut $\Theta(w)$ by hyperplanes and lower its dimension, which is how phasing information enters [8]. Minimum flow decomposition is treated only to the extent of Section 3.

Provenance. Splicing graphs are [7]; the non-identifiability of isoform deconvolution is [10, 8]; the $|E| - |V| + 2$ bound on the number of paths in a decomposition and the greedy algorithms are [14]; NP-hardness of minimum flow decomposition is [6]; practical exact solvers are [9, 5]; the Dscam isoform count is [12]; the BEST theorem is [1, 13] and its per-sequence normalisation is [3]. The tropical geometry of statistical models, which the planning of this note once proposed to bring in, is [11] and is not used: nothing in the polytope $\Theta(w)$ is tropical. We are not aware of prior work stating the dimension of the isoform fibre in closed form, or of the BEST identity of Theorem 7; a targeted search found neither.

Open. *First*, whether the integer minimum can exceed the real minimum for an integer flow on a DAG. *Second*, a formula or complexity classification for the number of integer points of $\Theta(w)$, as opposed to the multinomial-weighted sum of Theorem 7. *Third*, the effect of multi-junction reads on the dimension: each read spanning a set J of junctions with allele pattern π adds the constraint that the total abundance of isoforms with pattern π on J be the observed count, and the dimension after all such constraints is the true residual ambiguity of a sequencing design. *Fourth*, the analogue of Theorem 7 with the return arcs weighted, which would give a generating function in F rather than a single number.

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